

PTZrobot OTA Firmware Update Utility

User Instruction Manual (Linux PC)

Overview

This utility automates the process of updating PTZrobot firmware devices using OTA (Over-The-Air) updates.

The script will:

1. Download the latest firmware package from GitHub.
2. Extract the OTA firmware files.
3. Start a temporary HTTP server on your Linux PC.
4. Instruct the gateway to fetch and install the firmware.
5. Display transfer progress during the update.

Requirements

Before using the OTA updater, ensure the following:

Software Requirements

- Linux (Ubuntu, Debian, Mint, Fedora, etc.)
- Python 3.x installed
- Internet access

Network Requirements

- The Linux PC and PTZrobot gateway must be on the same network.
- The gateway must be able to reach the Linux PC.

Example: Linux PC 192.168.1.50 Gateway 192.168.1.100

First-Time Setup

The next 3 steps are only required the first time you use the updater. In the future this will all be wrapped up in a simple one step installer app which checks to see if python is already installed on the users system and if not install it before going on but for now.

Step 1: Verify Python Installation

Open a terminal window::

Type: `python3 --version`

If python is already installed you should see: `: Python 3.x.x`

If Python is not installed::

- Type: `sudo apt update` press Enter
- Type: `sudo apt install python3 python3-pip` press Enter

Step 2: Install Required Python Module

Install the Requests library:

- Type: `pip3 install requests` or `python3 -m pip install requests` press Enter

To Verify installation:

- Type: `python3 -c "import requests; print(requests.__version__)"` press |Enter

Step 3: Create an OTA Folder

Create a dedicated folder:for holing all your update files and the update utility

- Type: mkdir ~/PTZrobotOTA Press Enter
- Type: cd ~/PTZrobotOTA press Enter

You are now in the new folder you created Copy the updater script into this folder:

From your browser copy the update utility onto your system from here:

<https://github.com/digitalbird01/PTZrobot/blob/main/OTAbin/FWupdate.py>

Then move or copy the file [FWupdate.py](#) into the folder you just created.

That's it you will not have to perform that task again.

During the Update **Do not:**

- Close the terminal.
- Stop the Python script.
- Shut down the Linux system.
- Disconnect the network.

The HTTP server must remain running until firmware transfer completes

Running an OTA Update

Step 1: Open a Terminal window

Navigate to the OTA folder:you created in step 3 above

- Type: cd ~/PTZrobotOTA press Enter

Step 2: Start the Updater

- Type: python3 FWupdate.py Press Enter

Step 3: Enter the Gateway IP Address

You will be prompted for the device IP address which will be displayed on the device OLED

Example: 192.168.1.100

Automatic Update Process

1. Browser Opens Automatically

The script will attempt to open your default web browser and auto download:

<https://github.com/digitalbird01/PTZrobot/archive/refs/heads/main.zip>

This file contains all our current PTZrobot binaries

2. Firmware Package Detection

- The script monitors your Downloads folder and waits for PTZrobot-main.zip to arrive. You should see:

[UPDATE] Waiting for PTZrobot-main.zip to appear in Downloads... on your terminal window

- Once found you should see:

[UPDATE] Found ZIP:

/home/username/Downloads/PTZrobot-main.zip

3. Firmware Extraction

The updater extracts files from the repository's: into your updates folder:

Example output:

[UPDATE] Removed old file: manifest.json

[UPDATE] Removed old file: gateway.bin

[UPDATE] Saved manifest.json

[UPDATE] Saved gateway.bin

4. Local Firmware Server auto starts

The script starts an HTTP server on port 8000

Example output:

[UPDATE] Serving firmware on port 8000

[UPDATE] HTTP server is running and ready.

5. Gateway Update Triggered

The updater sends the current manifest to the device gateway

The device gateway looks at the manifest and decides if there is a valid update. If the device is already up to date the utility ends here. If the device sees a valid update the process continues to step6

6. Firmware Download Begins

As the gateway downloads firmware from your Linux PC, progress is displayed:

Example:

[SERVER] Sending file (1946254 bytes)

[SERVER] 65536/1946254 bytes (3.4%)

[SERVER] 131072/1946254 bytes (6.8%)

[SERVER] Transfer complete

When you see : [SERVER] Transfer complete Allow the device time to finish programming and reboot.. Then simply close the terminal window or stop the updater by pressing Ctrl + C

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Troubleshooting

Gateway does not update

Check:

- Correct gateway IP entered.
 - Linux PC and gateway are on the same subnet.
 - Firewall allows TCP port 8000.
 - The gateway can reach the Linux PC.
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ZIP file not detected

Verify the ZIP file exists in:

~/Downloads

and is named: PTZrobot-main.zip

Connection refused

Confirm the HTTP server is listening:

Type: `ss -tlnp | grep 8000` press Enter

Expected output should show Python listening on port 8000.

Verify OTA Files Are Available

List extracted files:

Is -l ~/PTZrobotOTA

You should see items such as:

manifest.json

Gateway RevB_Vx.xx.xx.bin

FWupdate.py

Determine Local IP Address

If unsure which IP the gateway should use to reach your PC:

ip route get 8.8.8.8

or:

hostname -I

Linux Firewall Configuration

If a firewall is enabled, allow incoming TCP port 8000.

Ubuntu/Debian (UFW)

Check status: Type: `sudo ufw status` press Enter

Allow port 8000:

Type: `sudo ufw allow 8000/tcp` press Enter

Typical Linux OTA Session Over View

1. `cd ~/PTZrobotOTA`
2. `python3 ota_update.py 192.168.1.100`
3. Show more lines
4. Expected sequence:
5. Plain Text
6. [UPDATE] Opening browser for firmware download...
7. [UPDATE] Waiting for PTZrobot-main.zip to appear in Downloads...
8. [UPDATE] Found ZIP...
9. [UPDATE] Extracting...
10. [UPDATE] Serving firmware on port 8000
11. [UPDATE] HTTP server is running and ready.
12. [UPDATE] Triggering gateway update...
13. [UPDATE] Gateway response: 200
14. [SERVER] Sending file...
15. [SERVER] Transfer complete